

PAIM — Predictive Assumption Identification Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Assumption Standard (PAS)

Operational Layer: Prediction Assumption Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Predictive Assumption Identification

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Predictive Assumption Identification

Canonical Definition

Predictive Assumption Identification Module (PAIM) defines the governance conditions under which assumptions materially supporting a prediction are discovered, identified, explicitly represented, bounded, classified, and associated with the predictive context in which they operate.

PAIM transforms predictive premises that may otherwise remain implicit, hidden, ambiguous, or embedded within methodology, data interpretation, system configuration, human judgment, or expected future conditions into identifiable governed assumption objects.

Its foundational principle is:



A hidden assumption cannot remain a governed assumption.

PAIM does not determine detailed assumption dependency, sensitivity, materiality, or change governance.

Those functions belong to the subsequent PAS modules.

PAIM establishes the governed assumption objects upon which those functions operate.

Operational Role

PAIM serves as the first internal governance space of the Prediction Assumption Standard (PAS) .

It receives the governed predictive context established through:

Prediction Object & Purpose

Prediction Methodology

Prediction Basis

and identifies the assumptions required for the prediction to remain meaningful.

The module asks:

What is this prediction treating as holding, continuing, remaining applicable, not occurring, or otherwise being sufficiently true for the prediction to make sense?

The governed progression is:

Governed Predictive Context → Assumption Discovery → Assumption Identification → Explicit Representation → Assumption Bounding → Initial Assumption State → Identified Predictive Assumptions

The resulting assumptions become input to:

ADMM — Assumption Dependency Mapping Module.

Internal Module Flow

→ Predictive Assumption Identification → Assumption Dependency Mapping → Assumption Sensitivity → Assumption Materiality → Assumption Change Governance

Current Position: Predictive Assumption Identification

Module Operational Space

PAIM governs:

assumption discovery; assumption identification; explicit assumptions; implicit assumptions; hidden assumptions; assumption identity; assumption description; assumption scope; assumption boundary; assumption origin; assumption type; assumption subject; assumption condition; continuity assumptions; methodological assumptions; operational assumptions; behavioral assumptions; environmental assumptions; market assumptions; organizational assumptions; policy and regulatory assumptions; system assumptions; resource assumptions; temporal assumptions; boundary assumptions; relationship assumptions; no-event assumptions; human assumptions; machine and AI assumptions; assumption granularity; assumption ambiguity; compound assumptions; assumption normalization; assumption duplication; initial assumption state; assumption uncertainty; unsupported assumptions; assumption identification gaps; and historical identification state.

Module Function

Predictions often depend upon assumptions without explicitly declaring them.

A forecasting system may say:

Demand will increase by 12%.

But that result may silently depend upon assumptions such as:

prices remain within a defined range; customer behavior remains comparable; no major competitor enters the market; supply remains available; regulation remains unchanged; the historical relationship used by the model remains applicable.

Without identifying these assumptions, users see the prediction output but not the conditions underneath it.

PAIM creates the governance transition from:

Unstated Predictive Premises

to:

Identified Predictive Assumptions

Predictive Assumption

A Predictive Assumption is a proposition, condition, relationship, continuity expectation, state expectation, simplification, or other premise treated as holding sufficiently for the prediction or its interpretation to remain meaningful.

A Predictive Assumption may concern:

what continues

what remains unchanged

what is expected to happen

what is expected not to happen

which relationship remains applicable

which condition remains within a defined range

or:

which state is treated as holding.

Assumption Discovery

Assumption Discovery is the process of exposing premises upon which the prediction materially depends.

Potential discovery sources include:

Prediction Methodology; Prediction Basis; model structure; system configuration; human judgment; expert reasoning;

scenario definitions; operational conditions; historical patterns; boundary definitions; predictive limitations; organizational expectations.

The objective is not to generate every imaginable assumption.

It is to surface assumptions that genuinely participate in the predictive structure.

Explicit Assumption

An Explicit Assumption is already represented as an assumption within the predictive process.

For example:

Assumption: Interest rates remain between 3% and 4% throughout the Prediction Horizon.

PAIM records the assumption as an identifiable governed object.

Implicit Assumption

An Implicit Assumption exists where the prediction depends upon a condition without sufficiently representing it as an assumption.

For example, a model trained on historical consumer behavior may implicitly assume that:

the relevant behavioral relationship continues into the Prediction Horizon.

PAIM seeks to surface materially relevant implicit assumptions.

Implicit Assumption $\neq$ Governed Assumption

Hidden Assumption

A Hidden Assumption is a predictive premise that materially affects the prediction while remaining insufficiently visible to the governed predictive process.

Hidden assumptions may exist inside:

algorithms; model configurations; spreadsheets; analyst judgment; organizational convention; data transformations; scenarios; system integrations; inherited methodologies.

PAIM does not permit material assumptions to remain intentionally invisible within a governed assumption structure.

A hidden assumption cannot remain a governed assumption.

Assumption Identity

Each material assumption should possess sufficient identity to distinguish it from other assumptions.

Identity may include:

assumption identifier; prediction reference; assumption statement; assumption category; assumption scope; temporal applicability; origin; initial state.

Assumption identity enables later dependency, sensitivity, materiality, and change governance.

Assumption Statement

An assumption must be stated with enough clarity to determine what condition is actually being assumed.

Statements such as:

"Market conditions remain normal."

may be too ambiguous.

Where material, the assumption should instead establish what "normal" means within the predictive context.

For example:

"No market-wide disruption materially affecting demand occurs during the 30-day Prediction Horizon."

Assumption Scope

Assumption Scope identifies the part of the prediction to which an assumption applies.

It may apply to:

the entire prediction; one methodology component; one target; one time interval; one geography; one operating state; one scenario; one population; one source relationship; one predictive condition.

Assumption Presence $\neq$ Universal Applicability

Assumption Boundary

Assumption Boundary defines where the assumption applies and where it does not.

A prediction may assume stable regulation:

within one jurisdiction

without making the same assumption for every jurisdiction included in another predictive context.

Boundaries prevent assumptions from expanding beyond their legitimate scope.

Assumption Origin

Where material, PAIM identifies where an assumption originated.

Possible origins include:

methodology; model; human analyst; expert; organization; Prediction Basis interpretation; scenario design; operational expectation; system configuration; external requirement; inherited predictive practice.

Assumption Origin $\neq$ Assumption Validity

Knowing where an assumption originated does not establish that it is correct.

Methodological Assumption

A Methodological Assumption arises from the structure or applicability conditions of Prediction Methodology.

Examples may include assumptions about:

distribution; stationarity; independence; variable relationships; sampling; model applicability.

PMES governs the methodology.

PAIM identifies the assumption as it applies to the specific prediction.

Operational Assumption

An Operational Assumption concerns an expected operational condition.

Examples:

The production line remains operational.

The distribution network remains available.

The system continues operating within specified parameters.

PAIM identifies the assumption.

ORA™ retains governance of Operational Reality.

Continuity Assumption

A Continuity Assumption treats a materially relevant relationship or condition as continuing into the Prediction Horizon.

Examples:

consumer behavior remains comparable; historical demand relationships persist; market structure remains sufficiently stable;

system performance characteristics remain comparable.

Historical Stability $\neq$ Future Stability

Behavioral Assumption

A Behavioral Assumption concerns expected behavior of:

individuals; populations; organizations; market participants; operators; agents; other decision-making entities.

Past behavior does not automatically establish future behavior.

Environmental Assumption

Environmental Assumptions concern external conditions such as:

weather; physical environment; geopolitical environment; macroeconomic environment; infrastructure conditions.

These may materially affect the prediction even where they are outside predictor control.

Market Assumption

A Market Assumption may concern:

liquidity; demand; pricing environment; competition; market structure; participant behavior; volatility regime.

A market condition should not silently be assumed stable where the prediction materially depends upon it.

Policy and Regulatory Assumption

A prediction may assume that relevant:

law; regulation; policy; licensing condition; administrative rule
remains sufficiently stable during the Prediction Horizon.

PAIM identifies the predictive assumption.

It does not determine the legal state itself.

System Assumption

A System Assumption concerns expected technical or operational system conditions.

Examples:

infrastructure remains available; system configuration remains unchanged; sensors remain functional; interfaces continue supplying information; required services remain accessible.

Resource Assumption

Predictions may depend upon continuing availability of:

capital; personnel; energy; computing resources; materials; production capacity; logistics capacity.

Where material, these dependencies should become explicit assumptions.

Temporal Assumption

A Temporal Assumption is bounded by time.

For example:

The condition is expected to remain stable for seven days.

The same assumption may not legitimately extend to twelve months.

Assumption Validity at Horizon A $\neq$ Assumption Validity at Horizon B

Boundary Assumption

A prediction may assume that the defined Predictive Boundary remains sufficiently stable.

This may concern:

geography; population; system scope; organizational structure; market scope.

Relationship Assumption

A Relationship Assumption treats a relationship between variables, states, events, or conditions as continuing to hold sufficiently for predictive use.

For example:

Increased mobility continues to correspond with increased retail demand.

SRVM may establish the signal's relevance.

PAIM identifies the assumption that the relationship continues.

No-Event Assumption

Some predictions depend upon something not happening.

Examples:

No major competitor enters the market.

No catastrophic infrastructure failure occurs.

No emergency regulation materially changes the operating environment.

Absence can therefore constitute a predictive assumption.

Human Assumption

A prediction may rely upon assumptions about:

expert judgment; operator behavior; management decisions; customer response; human compliance; human intervention.

Human behavior should not silently be treated as deterministic.

AI and Autonomous-System Assumptions

Predictions involving AI or autonomous systems may assume that:

model behavior remains sufficiently stable; system permissions remain unchanged; agents continue operating within defined boundaries; automated decision policies remain applicable; upstream models continue producing comparable outputs.

PAIM identifies these assumptions without absorbing AIG® or ASGA™ governance.

Assumption Granularity

An assumption should be represented at sufficient granularity to support meaningful governance.

Too broad:

"The economy remains okay."

Potentially governable:

"No recessionary condition materially reducing sector demand emerges during the Prediction Horizon."

Excessive fragmentation should also be avoided.

Assumption Governance ≠ Maximum Assumption Fragmentation

Compound Assumptions

A statement may contain several assumptions.

For example:

"The supplier remains operational, maintains current capacity, and continues delivering on schedule."

These may represent three materially different assumptions.

PAIM separates compound assumptions where their independent governance is necessary.

Assumption Normalization

Different statements may describe substantially the same assumption.

PAIM may normalize them into a consistent governed representation while preserving material distinctions.

This supports machine readability and prevents unnecessary duplication.

Duplicate Assumptions

The same assumption may appear in:

methodology documentation; model configuration; analyst notes; scenario descriptions.

Multiple representations do not automatically create multiple assumptions.

Multiple Assumption Statements $\neq$ Multiple Independent Assumptions

Initial Assumption State

At identification, an assumption may receive an initial state such as:

Established for Predictive Use

Holding

Conditionally Holding

Uncertain

Under Challenge

Unable to Determine

The state reflects the predictive governance condition.

It does not convert the assumption into fact.

Assumption Uncertainty

PAIM preserves uncertainty where the condition underlying an assumption cannot be established with confidence.

Forcing uncertain assumptions into binary states may create false certainty.

Assumption Identification $\neq$ Assumption Certainty

Supported Assumption

Prediction Basis may contain information supporting an assumption.

For example:

historical financial records may support an assumption that a supplier remains operational.

The assumption nevertheless remains distinct from the supporting information.

Supported Assumption $\neq$ Established Fact

Unsupported Assumption

An assumption may be identified even when its informational support is weak or unavailable.

Identification must not depend upon pretending support exists.

Unsupported assumptions should remain visible for later sensitivity and materiality governance.

Assumption Contradiction

Prediction Basis may contain information contradicting a proposed assumption.

PAIM should preserve the contradiction as part of the assumption's initial state or identification context.

It should not silently remove the assumption merely because it is inconvenient.

Assumption Identification Gap

An Assumption Identification Gap exists where a potentially material predictive premise cannot be sufficiently identified, expressed, bounded, or distinguished.

Examples include:

undocumented model assumption; unknown human judgment; ambiguous scenario condition; inherited methodology with unclear premises; unexplained system behavior.

Such gaps should remain explicit.

Historical Identification State

The assumptions identified when a prediction is formed must remain historically distinguishable from assumptions discovered later.

A later review may reveal an assumption that was hidden at formation time.

The record should distinguish:

Assumption Identified at Formation

from:

Assumption Discovered Retrospectively

Later Discovery $\neq$ Original Assumption Disclosure

This distinction becomes essential for historical prediction integrity.

Assumption Identification Sufficiency

Predictive Assumption Identification is sufficiently governed when:

material assumptions are identifiable; assumptions are explicitly stated; assumption scope is sufficiently bounded; origin is identifiable where material; implicit material assumptions are surfaced where reasonably possible; hidden assumptions become visible where discovered; compound assumptions are separated where necessary; duplicate representations are distinguishable; uncertainty remains visible; identification gaps remain explicit; historical identification state remains reconstructable.

Governance Requirements

PAIM requires that:

1. material predictive assumptions be identified; 2. assumptions be represented explicitly; 3. assumption statements possess sufficient semantic clarity; 4. assumption scope be bounded where material; 5. assumption origin be identified where relevant; 6. implicit material assumptions be surfaced where reasonably identifiable; 7. hidden assumptions become governed objects when discovered; 8. assumptions remain distinguishable from established facts; 9. assumptions remain distinguishable from Prediction Inputs; 10. assumptions remain distinguishable from predictions; 11. methodological assumptions be identifiable at prediction level where material; 12. operational and continuity assumptions remain explicit where material; 13. no-event assumptions be identifiable; 14. temporal applicability be represented where necessary; 15. compound assumptions be separated where independent governance is required; 16. duplicate representations not be mistaken for independent assumptions; 17. assumption uncertainty remain visible; 18. unsupported assumptions remain identifiable; 19. material contradictions remain visible; 20. Assumption Identification Gaps remain explicit; 21. historical identification state remain reconstructable.

Governance Logic

PAIM applies the following governance logic:

Governed Prediction Object & Purpose + Governed Prediction Methodology + Governed Prediction Basis

↓

Potential Predictive Premises Examined

↓

Explicit Assumptions Identified

↓

Implicit Assumptions Searched

↓

Material Hidden Assumptions Surfaced where identifiable

↓

Assumption Statements Defined

↓

Scope / Boundary / Temporal Applicability Established

↓

Assumption Origin Identified where material

↓

Compound / Duplicate Assumptions Resolved

↓

Initial Assumption State Established

↓

Uncertainty / Contradiction / Identification Gaps Preserved

↓

Identified Predictive Assumptions

↓

ADMM — Assumption Dependency Mapping Module

Where identification remains materially unresolved:

Assumption Identification Gap → Clarification / Disclosure / Decomposition / Additional Analysis / Explicit Uncertainty

Minimum Implementation Framework

A minimum implementation of PAIM must establish:

A. Prediction Reference

The governed prediction or predictive frame to which the assumption belongs.

B. Assumption Identifier

A sufficiently unique identifier for the assumption.

C. Assumption Statement

A clear representation of what is being assumed.

D. Assumption Type

Where applicable:

Methodological / Operational / Continuity / Behavioral / Environmental / Market / Regulatory / System / Resource / Temporal / Boundary / Relationship / No-Event / Human / AI-System / Other

E. Assumption Origin

Where materially identifiable.

F. Assumption Scope

The part of the predictive context to which the assumption applies.

G. Temporal Applicability

The period or Prediction Horizon over which the assumption is treated as applicable.

H. Explicitness State

At minimum:

Explicit / Surfaced Implicit / Previously Hidden / Retrospectively Discovered

I. Initial Assumption State

At minimum:

Established for Predictive Use / Holding / Conditionally Holding / Uncertain / Under Challenge / Unable to Determine

J. Support Reference

Where applicable, relevant Prediction Basis information supporting the assumption.

K. Contradiction Reference

Material information challenging the assumption.

L. Identification Gap

Any unresolved ambiguity, hidden premise, or incomplete assumption definition.

M. Historical Identification State

Whether the assumption was known at prediction formation or discovered later.

Governance Outputs

PAIM may produce:

Identified Predictive Assumptions; Predictive Assumption Register; Assumption Identifier; Assumption Statement; Assumption Type; Assumption Origin Record; Assumption Scope Record; Temporal Applicability Record; Explicitness State; Initial Assumption State; Assumption Support Reference;

Assumption Contradiction Record; Hidden Assumption Disclosure; Compound Assumption Decomposition; Duplicate Assumption Relationship; Assumption Identification Gap Register; Historical Assumption Identification Record.

The primary governed output is:

Identified Predictive Assumptions

This becomes the principal input to:

ADMM — Assumption Dependency Mapping Module.

Critical Distinctions

Prediction Assumption $\neq$ Established Fact

Implicit Assumption $\neq$ Governed Assumption

Hidden Assumption $\neq$ Governed Assumption

Prediction Input $\neq$ Prediction Assumption

Prediction Assumption $\neq$ Prediction

Assumption Presence $\neq$ Universal Applicability

Assumption Origin $\neq$ Assumption Validity

Historical Stability $\neq$ Future Stability

Assumption Validity at Horizon A $\neq$ Assumption Validity at Horizon B

Multiple Assumption Statements $\neq$ Multiple Independent Assumptions

Assumption Governance $\neq$ Maximum Assumption Fragmentation

Assumption Identification $\neq$ Assumption Certainty

Supported Assumption $\neq$ Established Fact

Later Discovery $\neq$ Original Assumption Disclosure

Enterprise Application

PAIM enables organizations to expose predictive dependencies that are often hidden behind apparently simple prediction outputs.

This is particularly important in:

AI forecasting + financial prediction + industrial systems + strategic planning + supply chains + healthcare + insurance + market forecasting + public policy + autonomous systems.

A prediction may be technically reproducible while still being methodologically opaque because nobody can explain what conditions it assumes will remain true.

PAIM creates the first governed assumption layer:

Prediction Output

is no longer interpreted without:

Predictive Conditions Assumed to Hold

This enables subsequent PAS modules to determine how those assumptions interact, how sensitive the prediction is to them, which are material, and what happens when they change.

Use Case 1 — The Forecast Assumes the Supplier Will Survive, but Nobody Wrote It Down

A manufacturer predicts that production targets will be achieved over the next six months.

Its forecasting system contains extensive information about:

demand; production capacity; delivery history; inventory; component requirements.

The primary supplier has historically delivered reliably.

No explicit assumption concerning supplier continuity appears in the forecast.

Yet the production prediction cannot realistically hold if that supplier stops operating.

The critical governance question is:

Can a prediction be considered fully governed when its success materially depends upon a condition that has never been identified as part of the predictive structure?

PAIM identifies:

Primary supplier remains operational and capable of supplying the required components during the Prediction Horizon

as a Predictive Assumption.

The assumption can then proceed to dependency, sensitivity, and materiality governance.

A hidden assumption cannot remain a governed assumption.

Use Case 2 — The AI Model Assumes the Future Behaves Like Its Training History

An AI system predicts customer demand using five years of historical transaction data.

The methodology is documented.

The Prediction Inputs are identified.

The sources are mapped.

The signals are relevant.

The Prediction Basis is sufficient.

However, the system does not explicitly represent one fundamental predictive premise:

the historical relationship between customer behavior, price, seasonality, and demand remains sufficiently applicable during the Prediction Horizon.

A new market structure begins emerging.

The system continues forecasting without acknowledging that its prediction depends upon continuity between historical and future behavior.

The critical governance question is:

Can a prediction remain methodologically transparent when the continuity relationship required for its historical information to remain predictive has never been identified as an assumption?

PAIM surfaces the continuity assumption as a governed predictive object.

Later PAS modules can determine its dependencies, sensitivity, materiality, and change conditions.

Canonical Boundary

PAIM governs the discovery, identification, explicit representation, classification, scope, boundary, origin, initial state, uncertainty, and historical identification status of assumptions materially supporting a prediction. It does not govern detailed Assumption Dependency Mapping, Assumption Sensitivity, Assumption Materiality, Assumption Change Governance, Prediction Basis, evidence validity, Prediction Formation, probability representation, feasibility, Operational Reality, or prediction-level revision.

PAIM establishes:

what the prediction is assuming.

It does not determine:

how important each assumption is, how strongly the prediction depends upon it, or what must happen when it changes.

Canonical Closing Statement

A prediction cannot be fully understood while the conditions it silently depends upon remain invisible.

PAIM turns predictive premises into identifiable governed assumptions before their dependencies, sensitivity, materiality, and change can be governed.

A hidden assumption cannot remain a governed assumption.

It

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